

Respiratory Acidosis, Respiratory Alkalosis, and Mixed Acid-Base Disorders

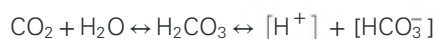
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RESPIRATORY ACIDOSIS

Respiratory acidosis is defined by a decrease in arterial pH that is primarily because of an increase in whole body carbon dioxide (CO₂) stores, which are indirectly assessed by the partial pressure of CO₂ in arterial blood (Paco₂). The Paco₂ is the same as the Pco₂ in alveolar air but is usually lower than in the capillaries (which reflects the Pco₂ in both the interstitial fluid and in the cells surrounding these capillaries).¹ At high altitude (i.e., above 4000 m or 13,000 feet), the underlying adaptation with chronic respiratory alkalosis usually causes a slight lowering of the Paco₂ at baseline so a respiratory acidosis may be present even with apparently normal Paco₂ levels.² Pseudorespiratory alkalosis refers to the presence of arterial eucapnia (or hypocapnia) with severe venous hypercapnia in patients with profound circulatory failure with preserved respiratory function.^{3,4}

PATHOPHYSIOLOGY OF RESPIRATORY ACIDOSIS

The quantity of CO₂ in the body remains constant if the lungs can excrete the amount of CO₂ produced from metabolism. This is reflected by the amount of dissolved CO₂ and in the partial pressure of CO₂ (Pco₂) in equilibrium with it. Although CO₂ is not an acid per se, it becomes a proton (H⁺) donor when a portion of the dissolved CO₂ combines with water to form carbonic acid as follows:



For any acid, the tendency to donate H⁺ is defined by the following Henderson-Hasselbalch equation:

$$\text{pH} = \text{pKa} + \log_{10}(\text{A}^-/\text{AH})$$

where pKa is the acid dissociation constant [A⁻] and [AH] are the concentrations of dissociated base and undissociated acid, respectively. When simplified, the following relationship arises for carbonic acid:

$$\text{pH} \propto \frac{[\text{HCO}_3^-]}{[\text{Paco}_2]}$$

The following relationship determines the Paco₂, which is a function of alveolar ventilation (VA) and CO₂ production (VCO₂)

$$\text{Paco}_2 = (\text{VCO}_2 \times K)/\text{VA}$$

where *K* is a proportionality constant.

When the Paco₂ increases above its usual physiologic levels (i.e., above 40–45 mm Hg [5.33–6 kPa]), acidemia usually develops.

$$\frac{[\text{HCO}_3^-]}{\uparrow [\text{Paco}_2]} \text{ results in } \downarrow \text{pH}$$

A decrease in VA or an increase in VCO₂ can increase Paco₂ and cause respiratory acidosis. CO₂ is the major end product of oxidative metabolism of carbohydrates and fatty acids. Normally, the amount of CO₂ produced metabolically is approximately 10 mmol/min, corresponding to an acid load of 13,000 to 16,000 mmol/day. Titration of HCO₃⁻ by nonvolatile acids leads to additional production of CO₂, which can be substantially increased in certain acidoses (e.g., lactic acidosis in septic shock). CO₂ production can also be increased by up to 30% to 100% of this basal production in conditions associated with increased metabolism (e.g., vigorous exercise, increased caloric intake with parenteral nutrition, or increased catabolism, such as in burns, trauma, and infection).⁵ Similarly, low metabolic states such as coma, anesthesia, paralysis, or a switch to nonoxidative metabolism (e.g., with ketogenesis) lead to decreased CO₂ production.

Bicarbonate retention by the kidney in response to hypercapnia is slow; thus a sudden increase in Paco₂ leads to a rapid lowering of the systemic pH. Total minute ventilation (MV) is the sum of VA and dead space ventilation; hence either a decrease in MV or an increase in dead space ventilation can decrease VA. Dead space can be either anatomic or alveolar (the latter refers to the fraction of alveolar ventilation ventilating unperfused alveoli). Dead space ventilation increases in the following situations⁶:

1. When the alveolar-capillary interface is destroyed (e.g., emphysema)
2. When blood flow is reduced (i.e., low cardiac output)
3. When alveoli are overdistended (e.g., during positive-pressure ventilation)

The maintenance of eucapnia depends on the ability of the ventilator system to match VCO₂ by adjustment of the VA. Thus, respiratory acidosis (primary hypercapnia) results from an abnormality of either ventilatory effort (i.e., patients who will not breathe because of a lack of respiratory drive) and/or ventilatory output (i.e., patients who cannot breathe because of ventilator pump failure).¹ The causes of respiratory acidosis (acute or chronic) are listed in Table 15.1.

RESPONSE TO RESPIRATORY ACIDOSIS

The timing of different responses to respiratory acidosis is shown in Fig. 15.1. This includes buffering by intracellular and extracellular buffers and renal compensation.

BICARBONATE BUFFERING SYSTEM

Of the 15,000 to 30,000 mmol of protons delivered daily to the extracellular fluid (ECF), only 40 to 60 nmol/L are found free in the blood. The remainder are bound by the various buffer systems that are present in ECF and intracellular fluid (ICF), which can bind or release H⁺, thus preventing large changes in free H⁺ concentration. Quantitatively important buffers are the proteins present in the ECF (albumin and

TABLE 15.1 Causes of Respiratory Acidosis

Type of Defect	Mechanism or Type	Examples of Acute Etiologies	Examples of Chronic Etiologies
Depressed ventilatory drive	Congenital Acquired	Drug overdose (opiates, sedatives, alcohol), general anesthesia, cerebrovascular accident, head trauma, acute meningoencephalitis, hypothermia	Primary hypoventilation (Ondine's curse), obesity, hypoventilation syndrome, central sleep apnea, myxedema, brain tumor (primary or metastatic)
Neuromuscular disorders Spinal cord Peripheral nerves Neuromuscular junction	Trauma Vascular Tumors Demyelinating Other Phrenic nerve paralysis Autoimmune Iatrogenic Infections/toxins	Cervical spinal cord injury, vascular accident, Guillain-Barré syndrome, traumatic cardiac surgery, myasthenia gravis, neuromuscular blocking agents, botulism, tetanus, tick paralysis	Vascular accident, primary or metastatic tumor, amyotrophic lateral sclerosis, compression because of tumor, myasthenia gravis, multiple sclerosis, poliomyelitis
Ventilatory muscle weakness	Congenital Autoimmune Acquired	Hypokalemia, hypomagnesemia, hypophosphatemia, ciguatera, organophosphate and shellfish poisoning	Muscular dystrophy, poliomyelitis, dermatomyositis
Chest wall pleural diseases		Flail chest, right bandages, pneumothorax, hemothorax	Kyphoscoliosis, morbid obesity, abdominal distension from ascites/organomegaly, pleural effusion, malignancy, fibrosis
Upper airways Lower airways	Obstruction	Epiglottitis, foreign body, angioedema, acute exacerbation of COPD, acute severe asthma	Tumor, tracheomalacia, goiter, COPD
Lung disease (parenchymal or vascular)	High V/Q Low V/Q (shunt)	Shock, pulmonary embolism, acute exacerbation of COPD, pneumonia, pulmonary edema, atelectasis	Emphysema, interstitial lung disease
Miscellaneous	Increased CO ₂ production Exogenous CO ₂ inhalation	Fever, sepsis, burns, severe trauma, seizures, accident, rebreathing	

COPD, Chronic obstructive pulmonary disease; V/Q, ventilation/perfusion.

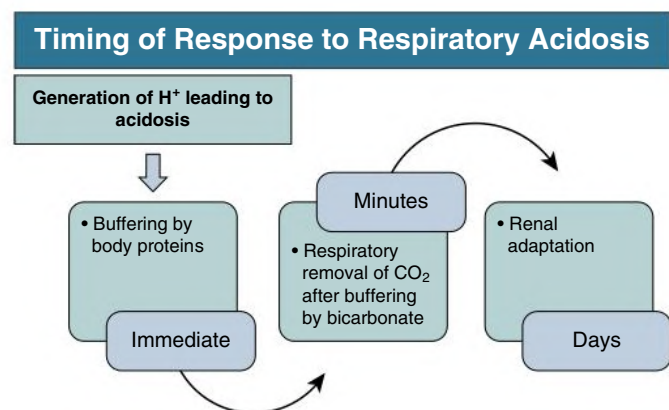
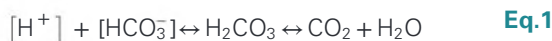


Fig. 15.1 Timing of response to respiratory acidosis.

immunoglobulins); bone; and amino acid side groups such as carboxylates, amines, and the histidines.

The main buffer is the bicarbonate buffering system that forms carbonic acid and then CO₂, as follows:



This process converts H⁺ released from a nonvolatile acid anion to CO₂, allowing it to be expelled from the ECF by the respiratory actions of the lungs. If CO₂ is effectively eliminated, this reaction continues from left to right. In fact, acidemia stimulates ventilation, further lowering the PaCO₂ and allowing more hydrogen ions to be buffered by bicarbonate. However, in some circumstances CO₂ accumulates at the tissue level, preventing the previous equation from going from left to right and may even drive the equation to the left. The

accumulation of CO₂ occurs when there is a rise in tissue (e.g., muscle) metabolic rate without a reciprocal increase in blood flow to the tissue. It may also occur when there is a decrease in tissue blood flow without a reciprocal decrease in the metabolic rate of the tissue.

KIDNEY RESPONSE TO RESPIRATORY ACIDOSIS

In respiratory acidosis, Equation 1 is driven to the left, leading to accumulation of H⁺ and acidemia. The overall response is shown in Fig. 15.2. At first, when CO₂ levels in the extracellular fluid (ECF) become elevated, HCO₃⁻ levels rise because of mass-action conversion of CO₂ to HCO₃⁻. This process is comparatively rapid (i.e., follows the onset of hypoventilation closely in time) and is not compensatory, but rather results in acidification of the plasma and the ECF because each mass action-produced HCO₃⁻ is also accompanied by one H⁺.

After the induction of acidosis, the kidney responds via the increased expression of H⁺ pumps in the apical membrane of tubular cells (largely in intercalated cells of the collecting duct). This process occurs over the course of several days, resulting in increased H⁺ secretion. Because at this early stage all filtered HCO₃⁻ has been reabsorbed proximally, these additional hydrogen ions are derived from CO₂ entering from the ECF (i.e., resulting in the synthesis of additional HCO₃⁻). More importantly, this secreted H⁺ is excreted in the urine.

However, this mechanism of compensation is ultimately limited by the fact that for each compensating H⁺ secreted, a HCO₃⁻ is synthesized. Also adding to the HCO₃⁻ pool is the bicarbonate formed by the original retention of CO₂, adding to the filtered load of HCO₃⁻. Thus, in the final phase of the renal response, most of the additional secreted H⁺ now goes to reabsorb this newly added HCO₃⁻, and the H⁺ excretion returns toward normal.

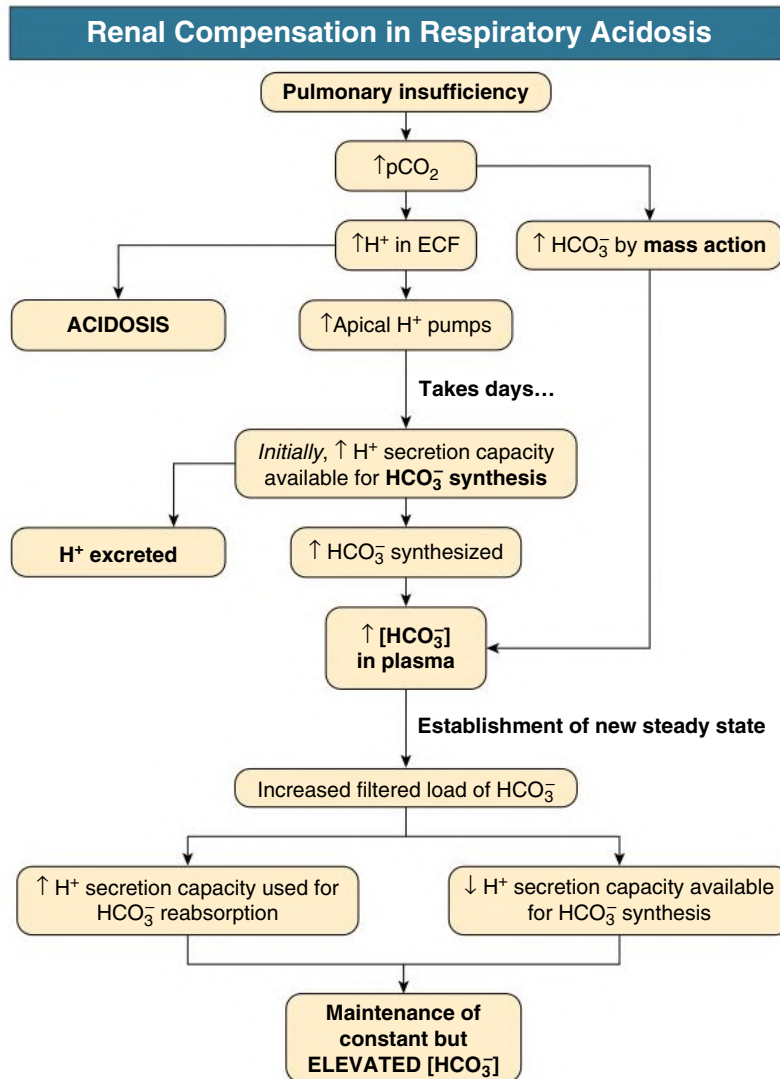


Fig. 15.2 Renal compensation in respiratory acidosis.

In summary, the kidneys initially respond to a respiratory acidosis by increasing the excretion of H^+ . Eventually, however, this compensation is limited by the increased filtered load of HCO_3^- . Thus, in the case of a prolonged respiratory acidosis, one generally sees a partial renal compensation with elevated HCO_3^- . Complete renal adaptation takes 3 to 5 days, and the expected change quantitatively is summarized in Table 15.2. The renal response to chronic hypercapnia is not altered by dietary sodium, chloride changes, or potassium changes, but in the presence of a chloride-deficient diet, recovery from chronic hypercapnia leads to a posthypercapnic metabolic alkalosis.⁷

Clinical Manifestations

Patients with acute failure of the ventilatory pump are usually dyspneic and tachypneic with overt respiratory distress, whereas patients with failure of ventilatory drive are usually hypoventilating or apneic. Hypercapnia has deleterious pulmonary and extrapulmonary consequences (Fig. 15.3). In the lungs, hypercapnia leads to pulmonary artery vasoconstriction, increases in right ventricular afterload, and right ventricular failure. In addition, hypercapnic acidosis may cause additional lung damage by increasing both nitric oxide production and inflammation and altering alveolar epithelial cells.⁸ Acute hypercapnia is associated with warm and flushed skin, sweating, and a bounding pulse because of increased cardiac output.⁹ This, however, can progress

to shock because of depressed cardiac output and decreased systemic vascular resistance in severe hypercapnia.¹⁰

Hypercapnia increases cerebral blood flow and thus causes increased intracranial pressure. This is usually manifested by dyspnea, anxiety, disorientation, headache, and irritability, which can progress to somnolence, confusion, and coma (hypercapnic encephalopathy). Physical findings can include abnormal involuntary movements, including myoclonus, asterix, tremors and, rarely, findings of increased intracranial pressure as evident by the finding of papilledema on retinal examination. Coma can occur in patients with acute exacerbations of conditions associated with chronic respiratory acidosis (e.g., chronic obstructive pulmonary disease [COPD]) because of treatment with high-flow oxygen.

Diagnosis

Whenever a respiratory acidosis is suspected, an arterial blood gas (ABG) should ideally be obtained to assess the overall acid-base status and pulmonary gas exchange. Appropriate compensatory responses should be assessed to rule out a component of respiratory acidosis in apparent metabolic disorders (see Table 15.2). However, an arterial puncture can be associated with complications, such as pain, infection, nerve injury, bleeding/hematoma, arterial aneurysm/pseudoaneurysm, dissection, thrombosis, and limb ischemia. Thus, there has been an increased use of venous blood gas (VBG) to replace frequent ABGs for diagnosis of

TABLE 15.2 Rules of Compensation in Acid-Base Disorders

Primary Acid-Base Disorder	Change in Laboratory Parameter	Compensatory Response	Expected Change in Laboratory Parameter with Compensatory Response
Metabolic acidosis	$\downarrow \text{HCO}_3^-$	Respiratory alkalosis	$\downarrow \text{Paco}_2$ 1–1.5 mm Hg for each mEq/L decrease in HCO_3^- or Expected $\text{Paco}_2 = 1.5 \times [\text{HCO}_3^-] + 8 \pm 2$
Metabolic alkalosis	$\uparrow \text{HCO}_3^-$	Respiratory acidosis	$\uparrow \text{Paco}_2$ 0.6–0.7 mm Hg for each mEq/L increase in HCO_3^-
Respiratory acidosis	$\uparrow \text{Paco}_2$	Metabolic alkalosis	$\uparrow \text{HCO}_3^-$ Acute: 1 mEq/L for each 10 mm Hg $\uparrow$ in Paco_2 Chronic: 4 mEq/L for each 10 mm Hg $\uparrow$ in Paco_2
Respiratory alkalosis	$\downarrow \text{Paco}_2$	Metabolic acidosis	$\downarrow \text{HCO}_3^-$ Acute: 2 mEq/L for each 10 mm Hg $\downarrow$ in Paco_2 Chronic: 5 mEq/L for each 10 mm Hg $\downarrow$ in Paco_2

Systemic Effects of Hypercapnia

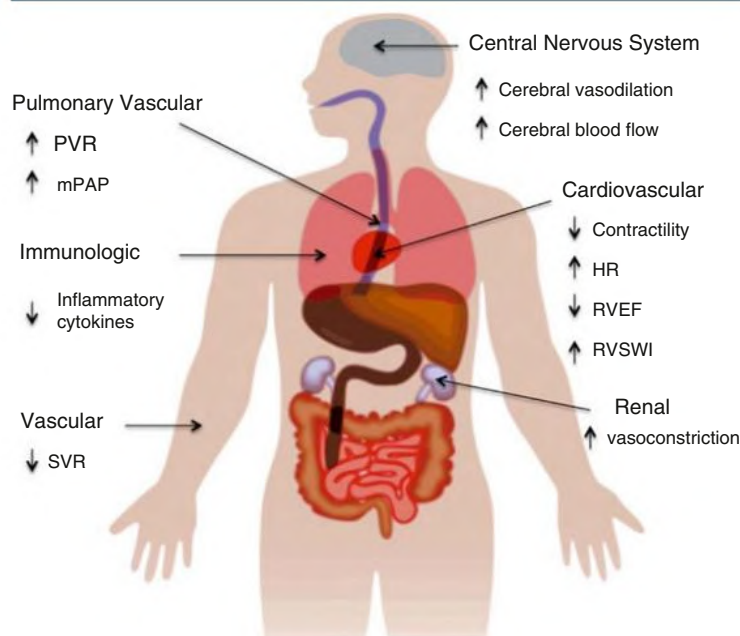


Fig. 15.3 Systemic effects of hypercapnia. HR, Heart rate; mPAP, mean pulmonary arterial pressure; PVR, pulmonary vascular resistance; RVEF, right ventricular ejection fraction; RVSWI, right ventricular stroke work index; SVR, systemic vascular resistance. (From Barnes T, Zochios V, Parhar K. Re-examining permissive hypercapnia in ARDS: A narrative review. *Chest*. 2018;154[1]:185–195.)

acid-base disorders in the emergency setting. Based on meta-analyses of studies looking at the correlation of results between ABG and VBG, venous and arterial pH and bicarbonate agree reasonably at all values, but the agreement is highest at normal values. However, peripheral venous CO_2 (Pvco_2) cannot be relied on as an absolute representation of Paco_2 . Despite this, a normal peripheral Pvco_2 has a good negative predictive value for normal arterial Pco_2 , and a normal Pvco_2 can be used as a screen to exclude hypercapnic respiratory disease.¹¹

The overall assessment of the ABG helps differentiate between acute and chronic respiratory acidosis. Acute respiratory acidosis is usually caused by an abrupt decline in ventilation that causes the Paco_2 to rise and pH to fall. In contrast, chronic respiratory acidosis is associated with elevated Paco_2 values but only slight decreases in pH because of effective renal compensation. Sometimes an acute respiratory acidosis can be superimposed on a chronic respiratory

acidosis. In this case, the change in the pH is not as much as would be expected for a similar acute change in the Paco_2 (in a person with baseline normal respiratory status). In this situation, the chronically elevated HCO_3^- reduces the fall in the pH and the kidney rapidly increases H^+ secretion when an acute rise in Paco_2 is superimposed on a chronic rise in Paco_2 .¹²

Hypercapnia is usually present along with hypoxemia because alveolar hypoventilation causes a proportional fall in alveolar oxygen pressure (PAO_2), according to the alveolar gas equation:

$$\text{PAO}_2 = \text{PiO}_2 - [\text{Paco}_2/R]$$

where arterial Pco_2 is assumed to be nearly the same as alveolar Pco_2 , PiO_2 is the inspired Po_2 (i.e., the inspired oxygen fraction [FiO_2] multiplied by the difference of barometric pressure and 47 [water vapor pressure at body temperature]), and R is the respiratory exchange ratio

(ratio between the amount of CO_2 produced in metabolism and O_2 used). At sea level with 21% FiO_2 , this can be simplified to the following:

$$\text{PAO}_2 = 150 - [1.25 \times \text{Paco}_2]$$

Calculating alveolar PO_2 using this relationship permits determination of the alveolar-arterial oxygen pressure difference (commonly called the “A–a gradient”). The normal A–a gradient is estimated by the following $(\text{Age} + 10)/4$ (i.e., ranges between 5 and 25 mm Hg). This calculation distinguishes between pure hypoventilation as an explanation for hypoxemia (in which case the A–a gradient is normal) and other mechanisms such as low ventilation-perfusion (V/Q) ratios and right-to-left shunt (in which case the A–a gradient is increased). The use of the alveolar gas equation is demonstrated in [Box 15.1](#).

Management

An overview of the management of acute respiratory acidosis is presented in [Fig. 15.4](#). The management of respiratory acidosis is dependent on the cause of the acidosis, and treatment of the underlying cause can often reverse the respiratory acidosis in acute ventilatory failure. However, supportive treatment is often necessary, especially when the etiology is not rapidly or fully reversible. Immediate treatment involves restoring the patency of the airway (if compromised) and delivery of

oxygen to maintain a PaO_2 of greater than 60 mm Hg and a peripheral oxygen saturation of greater than 90%. Several oxygen delivery devices are available ([Table 15.3](#)).

If appropriate oxygenation cannot be achieved with these oxygen delivery devices or if increased work of breathing is leading to impending respiratory fatigue, either noninvasive ventilation (NIV) or invasive mechanical ventilation can be used. NIV is delivered via tight-fitting face masks and is used for patients with acute respiratory failure who will benefit from ventilatory support but may not need conventional mechanical ventilation (which requires endotracheal intubation). Ideally, NIV is reserved for patients without altered mentation and who are hemodynamically stable. There are three modes of ventilation available for NIV: (1) continuous positive airway pressure (CPAP), (2) bilevel positive airway pressure (BiPAP), and (3) pressure support ventilation (PSV). The latter two modes of ventilation (i.e., BiPAP and PSV) are also referred to as noninvasive positive pressure ventilation (NIPPV). Endotracheal intubation is necessary when airway protection is paramount or with failure of NIV.

In patients with inadequate ventilatory drive despite a presumably normal ventilatory pump, management focuses on restoring normal alveolar ventilation. In the setting of acute central hypoventilation, invasive mechanical ventilation is usually more rapid and protects the airway reliably versus NIV. In chronic neuromuscular disorders, NIV is often used (e.g., elective initiation of noninvasive ventilation is becoming a standard of care in motor neuron disease patients with progressive ventilatory impairment).⁵

In acute respiratory distress syndrome (ARDS), normal tidal volumes can cause regional overdistension (so-called *volutrauma*) and repetitive opening and closing of alveoli can cause lung injury (sometimes called *atelectrauma*). Lung-protective ventilation using lower tidal volumes (less than 6 mL/kg predicted body weight) to maintain a plateau pressure less than 30 mm Hg prevents barotrauma and has been shown to reduce mortality.¹³ Because of this, the respiratory rate is set at a higher rate on the ventilator to maintain minute ventilation. This often requires a strategy of “permissive” hypercapnia, which

BOX 15.1 Examples Showing Use of the Alveolar Gas Equation				
At sea level, $\text{PAO}_2 = 150 - [1.25 \times \text{Paco}_2]$				
Normal A–a gradient = 5 to 25 mm Hg				
PaO_2	Paco_2	PAO_2	A–a Gradient	Possible Cause
70	40	100	30	V/Q mismatch
70	60	75	5	Hypoventilation

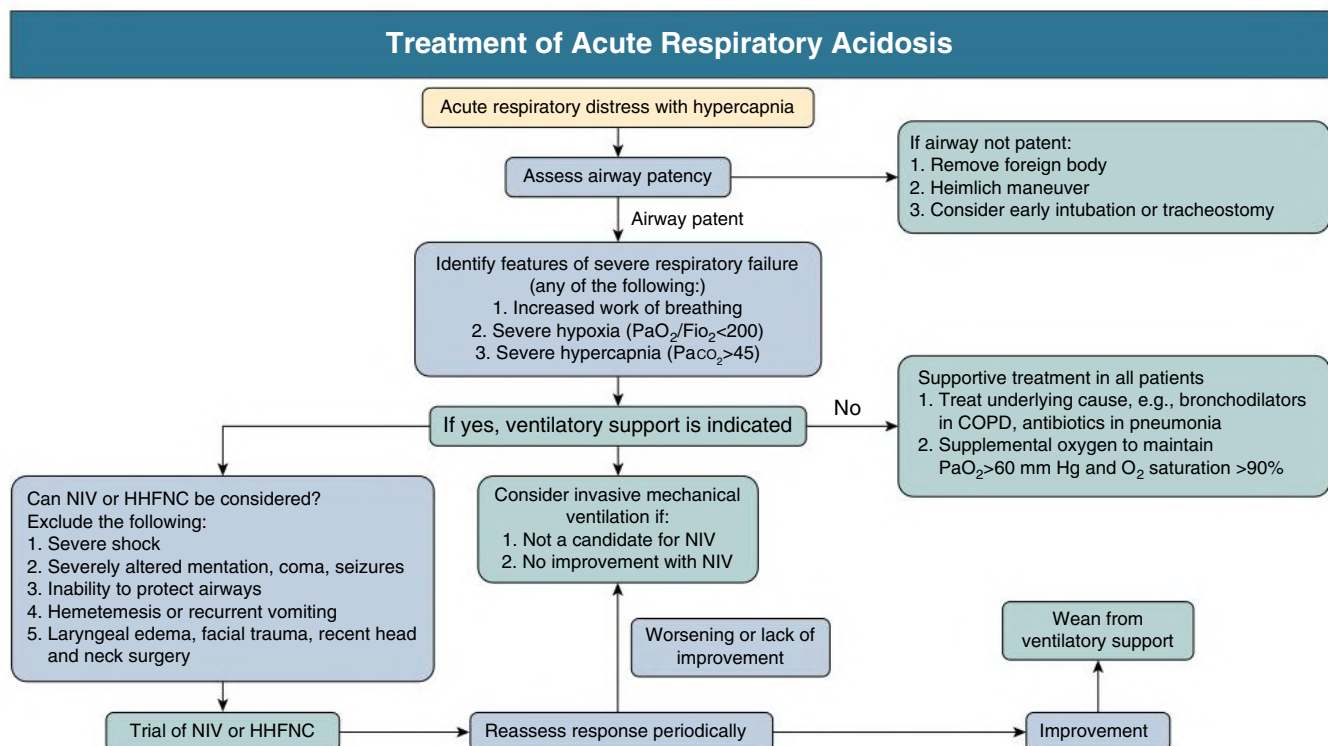


Fig. 15.4 Treatment of acute respiratory acidosis. HHFNC, Humidified high-flow nasal cannula; NIV, noninvasive ventilation.

TABLE 15.3 Oxygen Delivery Devices

System or Device	Flow Rates (L/min)	Reservoir Volume (mL)	F _i O ₂ (%)
Low-flow nasal canula	1–6	None	Each L/min adds 4% F _i O ₂ above room air
Standard face mask	6–12	100–200	35–60
Venturi mask	Fixed flow based on adapter	600–1000	24–40
Nonrebreather mask	>10	600–1000	80–100
High-flow nasal canula	40–60	None	30–100

can lead to a persistent respiratory acidosis with a pH of less than 7.3. In this setting, if the kidneys are not able to generate a compensatory metabolic alkalosis, bicarbonate infusions or renal replacement therapy have been used to keep the pH at least greater than 7.15. Detailed protocols are available to guide this ventilatory strategy.¹⁴ Although permissive hypercapnia was previously thought to be benign, it has several physiologic consequences, and alternative interventions, including prone ventilation, ultralow tidal volume ventilation, extracorporeal membrane oxygenation (ECMO), and extracorporeal CO₂ removal (ECCO₂R), are being used to minimize hypercapnia or even aim for normocapnia.⁸ Permissive hypercapnia can be dangerous in patients with traumatic brain injury or in those with cerebrovascular disease because it can trigger increased cerebral blood with increased intracranial pressure.

ECMO is reserved for patients with refractory severe ARDS (Pao₂:FiO₂ ratio < 60 mm Hg) after adequate lung-protective ventilation and correction of volume overload have failed to improve oxygenation.¹⁵

ECCO₂R is a novel extracorporeal technique whose objective is the decarboxylation of blood. This leads to correction of hypercapnia and resolution of respiratory acidosis. This therapy is being tested in clinical trials of acute severe exacerbations of COPD to prevent intubation, to minimize invasive mechanical ventilation, and to facilitate weaning from mechanical ventilation. Additional clinical scenarios where it may be useful are in ARDS and as a bridge to lung transplantation.¹⁶

It is often not possible to reverse the cause of chronic respiratory acidosis. Adequate airway drainage, relief of bronchospasm, treatment of pulmonary infections, and treatment of heart failure may lead to improvement in the hypercapnia. In some situations (e.g., in advanced COPD or chronic neuromuscular diseases causing ventilator pump failure), patients may require long-term use of NIV or invasive mechanical ventilation.^{17,18}

RESPIRATORY ALKALOSIS

In respiratory alkalosis, the arterial pH increases primarily because of a decrease in whole body CO₂ stores, which are indirectly assessed by the Paco₂. Respiratory alkalosis is diagnosed by the finding of a Paco₂ less than 35 mm Hg (4.7 kPa) in patients with an increased arterial pH (>7.45). Respiratory alkalosis is commonly associated with a compensatory metabolic acidosis. This relationship, as reflected by the simplified Henderson Hasselbalch equation, is as follows:

$$\frac{[\text{HCO}_3^-]}{\downarrow [\text{Paco}_2]} \text{ results in } \uparrow \text{pH}$$

Failure to measure the systemic pH commonly leads to mislabeling of patients with respiratory alkalosis as having a primary metabolic acidosis when only the HCO₃⁻ level as measured in the blood chemistries is reviewed. As discussed earlier, pseudorespiratory alkalosis refers to the presence of arterial eucapnia (or hypocapnia) with severe venous hypercapnia in patients with profound circulatory failure with preserved respiratory function. This can also be seen because of hyperventilation when such patients are placed on mechanical ventilation.

TABLE 15.4 Causes of Respiratory Alkalosis

Mechanism	Causes
Hypoxia	High altitude, cyanotic congenital heart disease, congestive heart failure, intrinsic lung disease, laryngospasm
Pulmonary receptor stimulation	Pneumonia, asthma, pulmonary edema, pulmonary embolism, pulmonary fibrosis
Drugs	Salicylates, alkaloids, theophylline, progesterone, catecholamines
Central nervous system stimulation	Primary hyperventilation syndrome, subarachnoid hemorrhage, cerebrovascular accident, pain, trauma
Miscellaneous	Pregnancy, cirrhosis, sepsis, fever, anxiety-hyperventilation syndrome

Modified from Halperin, ML, Goldstein MB, Kamel KS. *Fluid, Electrolyte, and Acid-Base Physiology: A Problem-Based Approach*, 4th ed. Elsevier; 2010:222–239.

PATHOPHYSIOLOGY

Respiratory alkalosis is the most common acid-base disorder but often is not considered a serious clinical problem even though the mortality rate in patients with respiratory alkalosis is higher than the elderly general population.¹⁹

Paco₂ is normally maintained in the normal range by control of the ventilatory drive. Chemoreceptors in the brain (central chemoreceptors) and in the carotid bodies (peripheral chemoreceptors) sense H⁺ in the blood and adjust the ventilation to regulate Paco₂ and pH. Respiratory alkalosis is usually induced by a process involving hyperventilation. This leads to the shifting of Equation 1 to the right, effectively reducing the H⁺ concentration and causing an increase in the systemic pH. The causes of respiratory alkalosis include central causes, hypoxemic causes (leading to stimulation of peripheral chemoreceptors), pulmonary causes (leading to stimulation of peripheral afferent receptors), drugs, and some miscellaneous causes (Table 15.4). Common conditions associated with respiratory alkalosis include pregnancy (in which progesterone-induced hyperventilation leads to a mild respiratory alkalosis most evident in the third trimester)²⁰; cirrhosis (characterized by both hypoxemia and stimulation of respiratory drive via progesterone, ammonia, vasoactive intestinal peptide, and glutamine); and iatrogenic hyperventilation in patients on mechanical ventilation.

Hyperventilation syndrome is a type of dysfunctional breathing in which an inappropriate increase in minute ventilation beyond metabolic needs leads to a respiratory alkalosis and is associated with a wide range of symptoms without a clear organic precipitant. It is caused by stress and anxiety, both of which act on the behavioral respiratory control system. The hyperventilation ceases during sleep, when the behavioral control system is inactive and only the metabolic system is controlling breathing. The diagnosis of hyperventilation syndrome should be a diagnosis of exclusion.

RESPONSE TO RESPIRATORY ALKALOSIS

An increase in the pH because of respiratory alkalosis can be acute or chronic. Excess HCO_3^- levels are initially buffered by release of H^+ from intracellular sources to reduce HCO_3^- levels. This buffering is complete in minutes and persists for at least 2 hours.²¹ Within 2 to 3 days, there is progressive decrease in renal H^+ secretion (which leads to reduced generation of new HCO_3^- to replace the amount used for buffering the daily acid load) and increased HCO_3^- secretion by reduced reclamation of filtered HCO_3^- , both of which lead to reductions in serum HCO_3^- aimed at restoring physiologic pH. Therefore, acute respiratory alkalosis is associated with high bicarbonate levels because there has not been sufficient time to lower the HCO_3^- levels, and chronic respiratory alkalosis is associated with low to normal HCO_3^- levels. The stimulus for renal compensation is not pH, but rather Pco_2 .²²

Clinical Manifestations

Because the primary cause of all respiratory alkaloses is hyperventilation, most patients present with dyspnea. The specific clinical features depend on its duration, its severity, and the underlying disease process.

Acute respiratory alkalosis can cause light-headedness, confusion, peripheral and circumoral paresthesias, cramps, and syncope. The mechanism of these symptoms is cerebral vasoconstriction causing decreased cerebral blood flow.²³ Carpopedal spasm (which can often be elicited as the Trousseau sign or the Chvostek sign) may occur in severe cases because of decreased levels of ionized calcium in the blood (because of change in the degree of protein binding with change in the pH). Additional physical examination findings may include tachycardia, tachypnea, diaphoresis, hypertension or hypotension, and altered mental status. Chronic respiratory alkalosis is usually asymptomatic and has no distinctive signs.

Cardiovascular effects of hypocapnia in healthy and alert patients are minimal but can be significant in patients who are anesthetized, critically ill, or receiving mechanical ventilation. Cardiac output and systemic blood pressure may fall in these passively hyperventilated patients because of the effects of sedation and positive-pressure ventilation on venous return, systemic vascular resistance, and heart rate. Cardiac rhythm disturbances can also occur because of leftward shift of the hemoglobin-oxygen dissociation curve causing tissue hypoxia.

High altitude illness, which is usually associated with acute respiratory alkalosis, includes a spectrum of disorders that are important to recognize. Acute mountain sickness (AMS) is diagnosed clinically based on the appearance of typical symptoms in a person who ascends to a high altitude (generally >2000 m). Symptoms include headache, fatigue, light-headedness, anorexia, nausea, vomiting, frequent awakening, and dyspnea with exertion. This can progress to high altitude cerebral edema (HACE), which is heralded by the onset of generalized neurologic symptoms.²⁴

Diagnosis

A suspicion of respiratory alkalosis based on clinical features should lead to the measurement of an ABG. Failure to measure the systemic pH commonly leads to mislabeling of patients with respiratory alkalosis as having a primary normal anion gap metabolic acidosis when only the HCO_3^- level as measured in the blood chemistries is reviewed. This is very common in pregnancy and cirrhosis, which are usually associated with a respiratory alkalosis without overt respiratory distress.

Treatment

The treatment of respiratory alkalosis is primarily directed at correcting the underlying cause. Respiratory alkalosis itself is rarely life threatening and correcting significant hypoxemia (if present) is more critical. If the Paco_2 is corrected rapidly in patients with chronic respiratory

alkalosis, metabolic acidosis may develop because of persistence of the renal compensatory drop in serum bicarbonate.

In mechanically ventilated patients who have respiratory alkalosis, the minute ventilation can be decreased by lowering the tidal volume and/or the respiratory rate. Inadequate sedation and pain control may contribute to respiratory alkalosis in patients breathing over the set ventilator rate.

Acute management of patients who are hyperventilating includes reassurance, removal of any stressors, and breathing retraining with a focus on abdominal (diaphragmatic) breathing. Rebreathing of carbon dioxide (CO_2) by breathing into a paper bag is not recommended because it can cause significant hypoxemia, especially in the presence of underlying respiratory or cardiovascular disease. If conservative treatment is not successful, small doses of short-acting benzodiazepines can be used to decrease hyperventilation. Long-term management requires identification and treatment of underlying conditions and precipitants and cognitive behavioral therapy.

MIXED ACID-BASE DISORDERS

Mixed acid-base disorders refer to the presence of more than one acid-base disorder concomitantly. Careful analysis of the ABG is essential to uncover the presence of a mixed acid-base disorder because they can be an important clue to the underlying etiology. An appropriate compensatory response to a primary acid-base disorder is not considered as indicative of a mixed acid-base disorder (see Table 15.2 for the expected compensatory response to the primary acid-base disorders). The effect of a mixed acid-base disorder on the systemic pH can either be additive or can help limit the change in pH (e.g., the presence of both a metabolic acidosis and respiratory acidosis leads to a more severe acidosis, whereas the presence of a concomitant metabolic acidosis and respiratory alkalosis could lead to a minimal change in systemic pH). Triple acid-base disorders can also be seen and involve a combination of two or more independent processes causing metabolic acidosis (leading to a combined high anion gap metabolic acidosis and normal anion gap metabolic acidosis) in the presence of a respiratory disorder.

Diagnosis

Using the following approach to acid-base analysis can help correctly identify mixed acid-base disorders:

- Step 1: Look at the pH to classify if the person is acidemic or alkalemic.
- Step 2: Determine whether the primary disorder is metabolic or respiratory.
- Step 3: If the primary disorder is respiratory, is it chronic or acute? If the primary disorder is metabolic, is the anion gap elevated?
- Step 4: Is the compensation for the primary disorder appropriate (using Table 15.2)? If not, a mixed acid-base disorder is present.
- Step 5: In cases with a high anion gap metabolic acidosis, is there an additional metabolic disorder?

A simplified approach to the identification of mixed acid-base disorders is presented in Fig. 15.5, and the common mixed acid-base disorders are discussed later.

METABOLIC ACIDOSIS AND RESPIRATORY ACIDOSIS

In the presence of metabolic acidosis, the expected compensatory respiratory response is assessed using the Winters formula ($\text{Expected } \text{Paco}_2 = 1.5 \times [\text{HCO}_3^-] + 8 \pm 2$). If the measured Paco_2 exceeds the expected Paco_2 by 5 mm Hg, this indicates a concomitant respiratory acidosis. Common clinical scenarios where this is seen include

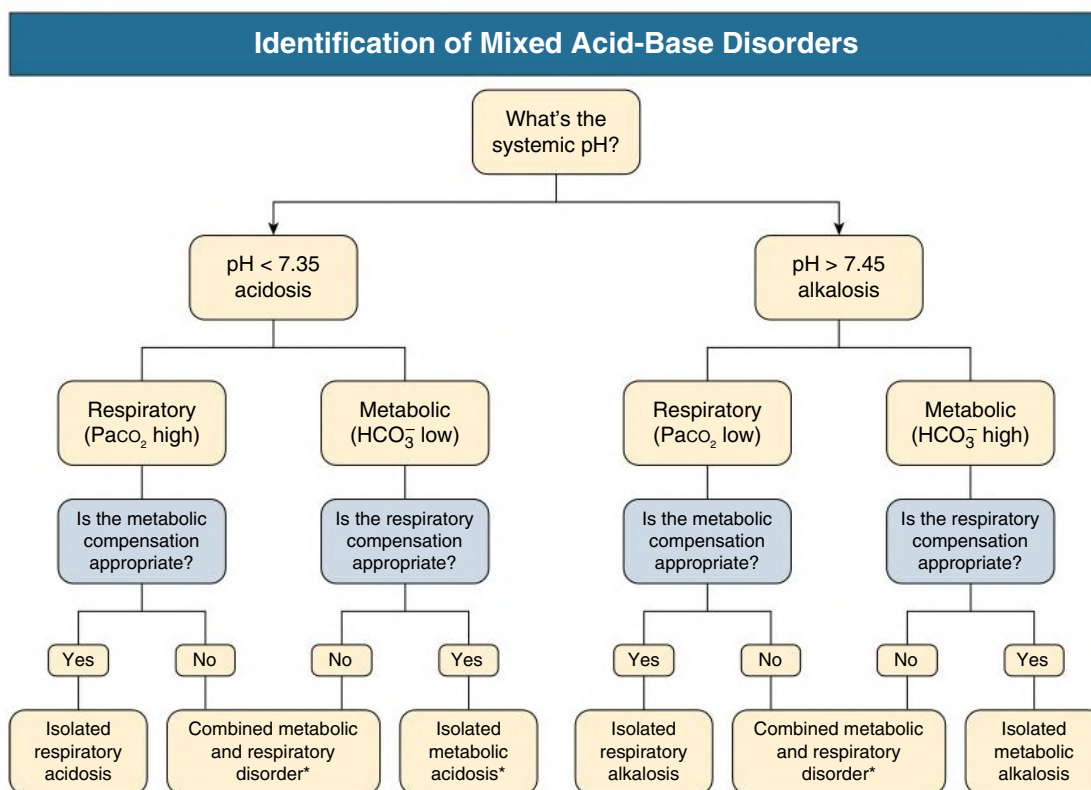


Fig. 15.5 Approach to identification of mixed acid-base disorders. *Evaluate for the presence of concomitant normal gap and high anion gap metabolic acidosis in every case of high anion gap metabolic acidosis.

cardiopulmonary arrest, severe kidney injury in patients with hypercapnic respiratory failure, shock with lactic acidosis in severe respiratory failure, respiratory muscle fatigue with septic shock, and in respiratory muscle weakness because of severe electrolyte disturbances (hypokalemia, hypophosphatemia) caused by prolonged diarrhea. The treatment should be focused on addressing the causes of the respiratory acidosis and the metabolic acidosis.

METABOLIC ACIDOSIS AND RESPIRATORY ALKALOSIS

In the presence of a metabolic acidosis, the expected compensatory respiratory response is assessed using the Winters formula (as above). If the measured Paco_2 is lower than this expected Paco_2 by 5 mm Hg, it indicates the presence of a concomitant respiratory alkalosis. Because of the opposing effects on systemic pH, the concomitant presence of these two disorders leads to a minimal change in pH. A classic condition associated with this pattern is salicylate intoxication, which leads to central stimulation of the ventilatory drive associated with a high anion gap metabolic acidosis because of a shift from oxidative phosphorylation to anaerobic metabolism and increased production of lactic acid and ketoacids. Thus, salicylate toxicity should always be ruled out in patients who present to the hospital with this mixed acid-base disorder. This combination is also commonly seen in critically ill patients with metabolic acidosis caused by lactic acidosis or severe kidney injury in combination with hyperventilation caused by fever, sepsis, hypoxia, and/or mechanical hyperventilation. Another common scenario is hepatic failure in which situation hyperventilation leads to respiratory alkalosis and metabolic acidosis is caused by lactic acidosis or acute kidney injury (AKI) or both.

METABOLIC ALKALOSIS AND RESPIRATORY ALKALOSIS

Metabolic alkalosis is associated with a compensatory respiratory acidosis. The expected increase in Paco_2 is 0.6 to 0.7 mm Hg for each mEq/L increase in HCO_3^- . If the measured Paco_2 is less than this expected Paco_2 , a concomitant respiratory alkalosis is present. A marked rise in the pH may result in this mixed acid-base disorder. Common conditions associated with respiratory alkalosis like cirrhosis, pregnancy, and mechanically hyperventilated patients can often develop a metabolic alkalosis in the setting of vomiting, prolonged nasogastric suction, diuretic use, severe hypokalemia, or with alkali administration (e.g., citrate in blood products, exogenous bicarbonate supplements).

METABOLIC ALKALOSIS AND RESPIRATORY ACIDOSIS

Metabolic alkalosis is associated with a compensatory respiratory acidosis. The expected increase in Paco_2 is 0.6 to 0.7 mm Hg for each mEq/L increase in HCO_3^- . If the measured Paco_2 is higher than this expected Paco_2 , a concomitant respiratory acidosis is present. Common scenarios that present with this mixed disorder include acute exacerbations of COPD with cor pulmonale, or decompensated heart failure leading to acute respiratory acidosis and then metabolic alkalosis because of aggressive diuresis. Alternatively, metabolic alkalosis with prolonged vomiting or diuretic use can be complicated by respiratory muscle paralysis because of severe electrolyte disturbances (hypokalemia, hypophosphatemia).

MIXED METABOLIC ACIDOSIS

The approach to the evaluation of a metabolic acidosis begins with evaluation of the anion gap. In the setting of a high anion gap (AG) acidosis, further characterization of the metabolic acid-base disorder is important. For this, the relationship between the increment in the anion gap (Δ AG) and the decrement in the serum bicarbonate concentration (Δ HCO₃) is reviewed. When the Δ HCO₃ is greater than the Δ AG, it suggests that a mixed high AG and hyperchloremic normal gap metabolic acidosis is present. Common causes of a mixed high AG and normal gap acidosis include severe AKI or chronic kidney disease, ketoacidosis or lactic acidosis with concomitant diarrhea, and recovery phase of diabetic ketoacidosis.

METABOLIC ACIDOSIS AND METABOLIC ALKALOSIS

As discussed previously, in the setting of a high AG acidosis, the Δ AG and the Δ HCO₃ are calculated to look for an additional metabolic disorder. When the Δ HCO₃ is less than the Δ AG, a mixed metabolic alkalosis and high AG metabolic acidosis is likely to be present. Common causes for this mixed disorder include metabolic alkalosis induced by vomiting or diuretic use in patients with diabetic ketoacidosis or lactic acidosis because of liver cirrhosis. Prolonged

nasogastric suction in patients with uremia or diabetic ketoacidosis can also present with the same problem.

TRIPLE ACID-BASE DISORDERS

Typical triple acid-base disorders involve the presence of a mixed metabolic disorder along with either a respiratory acidosis or a respiratory alkalosis. In the example previously cited, vomiting or diuretic use in patients with diabetic ketoacidosis or lactic acidosis because of liver cirrhosis can lead to a mixed metabolic acidosis and metabolic alkalosis—if such patients then develop a hypercapnic ventilatory failure, this will lead to a triple disorder with additional respiratory acidosis. Another scenario involves the presence of an acute or chronic respiratory disorder in combination with a metabolic disorder (e.g., a COPD patient with chronic respiratory acidosis who gets intubated and mechanically ventilated then develops a sudden decrease in PaCO₂). In the presence of an underlying compensatory metabolic alkalosis, a new metabolic alkalosis caused by vomiting or nasogastric suction could lead to a rapid elevation in the pH.

Treatment of Mixed Acid-Base Disorders

The management of mixed acid-base disorders involves treating the underlying cause of the acid-base disorder and treating individual components (i.e., each simple acid-base disorder). [Figs. 15.6 to 15.9](#)

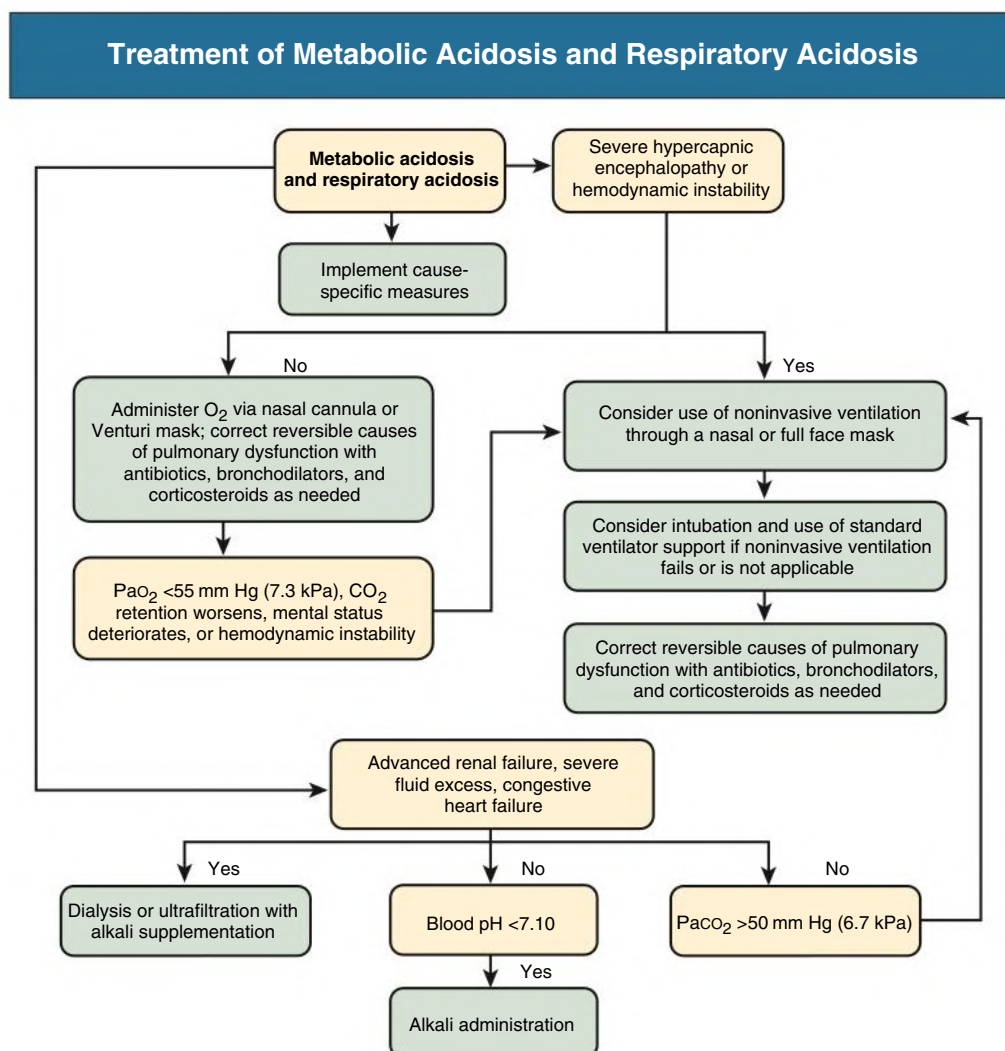


Fig. 15.6 Treatment of metabolic acidosis and respiratory acidosis.

Treatment of Metabolic Alkalosis and Respiratory Alkalosis

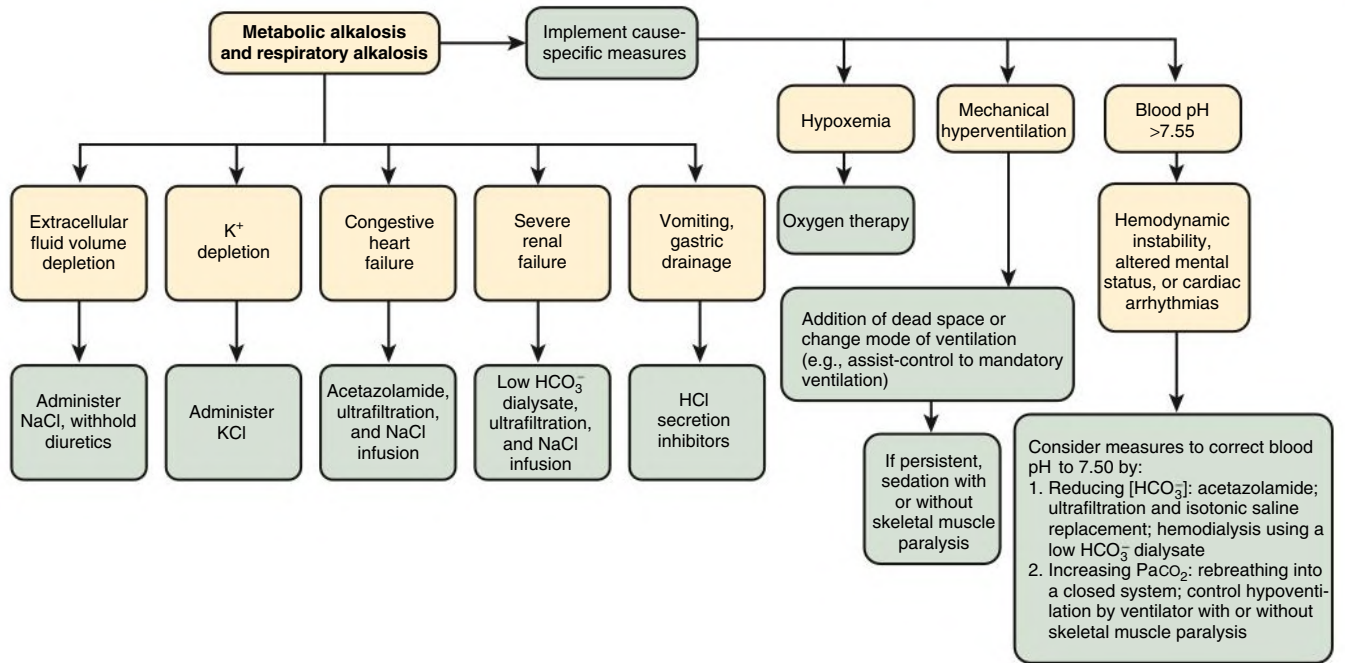


Fig. 15.7 Treatment of metabolic alkalosis and respiratory alkalosis.

Treatment of Metabolic Alkalosis and Respiratory Acidosis

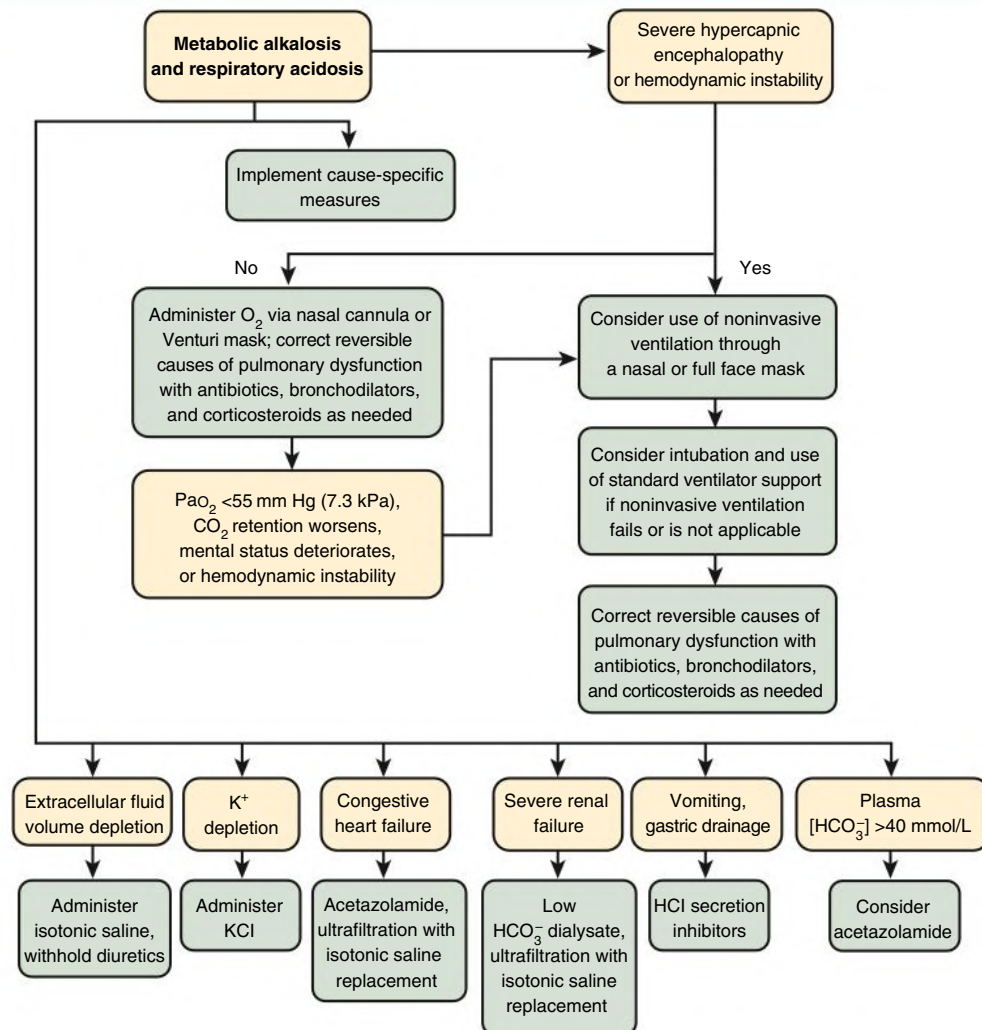


Fig. 15.8 Treatment of metabolic alkalosis and respiratory acidosis.

Treatment of Metabolic Acidosis and Respiratory Alkalosis

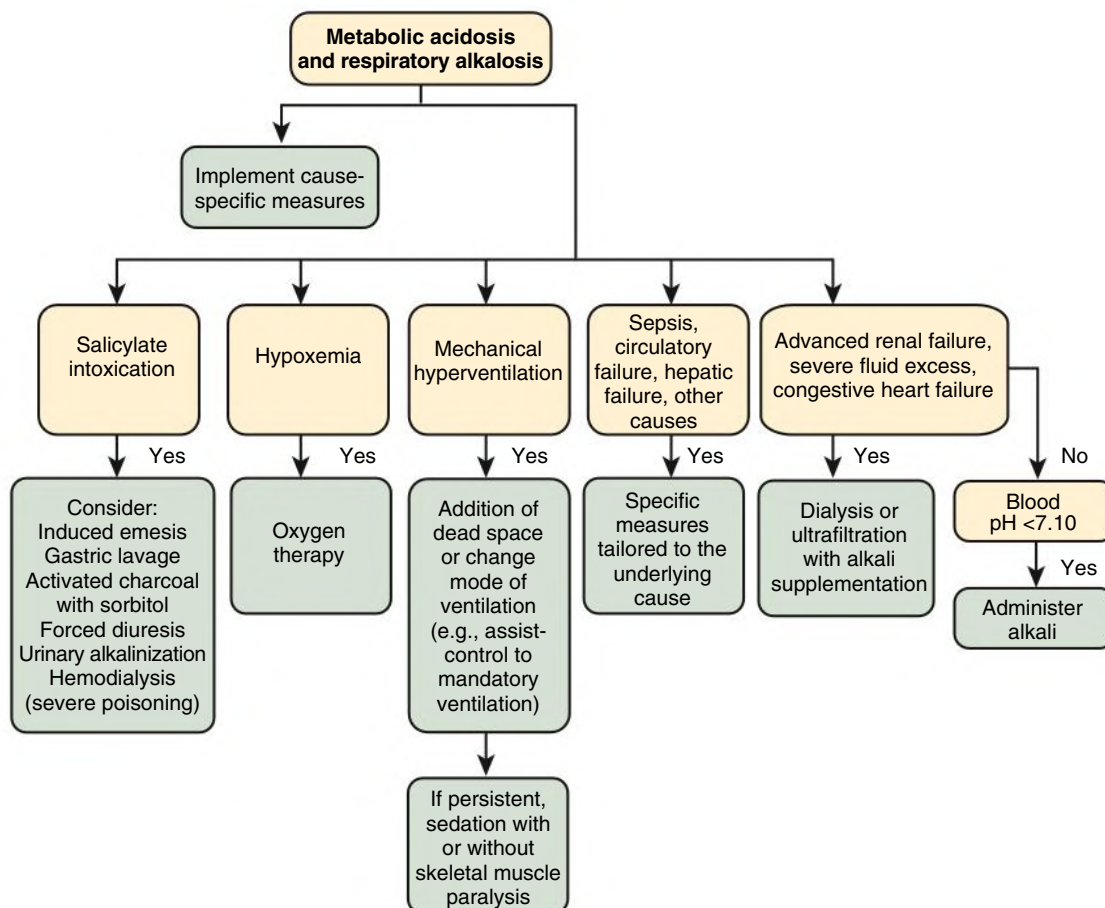


Fig. 15.9 Treatment of metabolic acidosis and respiratory alkalosis.

provide algorithms for the management of some of the more common mixed disorders. Certain disorders (e.g., combined metabolic and respiratory acidosis) lead to a severe change in the pH, and treatment of just one of the simple acid-base disorders can restore the pH to an acceptable level. On the other hand, disorders that have components

with opposing effects on the pH should be approached with caution (e.g., although a mixed metabolic acidosis and respiratory alkalosis may have a minimal effect on the pH, complete correction of one of the simple acid-base disorders could lead to a severe acid-base disorder).

SELF-ASSESSMENT QUESTIONS

1. A 26-year-old woman with asthma was brought to the emergency department for severe shortness of breath and appeared to be struggling for air. Her examination was notable for tachypnea with very shallow breaths. She had extensive wheezing throughout both lung fields and audible stridor. Her lab results are as follows:

	ARTERIAL			VENOUS			
pH	Pco ₂	Pao ₂	Na	K	Cl	HCO ₃ ⁻	
7.24	61	55	138	4.5	104	26	

- A. What acid-base disturbance is present?
 - B. Is compensation appropriate?
 - C. What is the underlying disorder that caused this problem?
2. A 77-year man is brought to the ICU after being found unresponsive. His initial lab results are as follows:

	ARTERIAL					VENOUS	
pH	Pco ₂	Pao ₂	Na	K	Cl	HCO ₃ ⁻	
7.07	28	59	135	2.5	100	8	

What is the acid-base disturbance?

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